

internal-external rotation 0 degree-abducted — ex vivo – in vivo difference, 95% CI (overlap only)

Sternoclavicular — Protraction(+)/retraction(-)

single group · K = 18

Sternoclavicular — Depression(+)/elevation(-)

single group · K = 17

Sternoclavicular — Backwards(+)/forward(-) rot

single group · K = 17

Acromioclavicular — Protraction(+)/retraction(-)

single group · K = 16

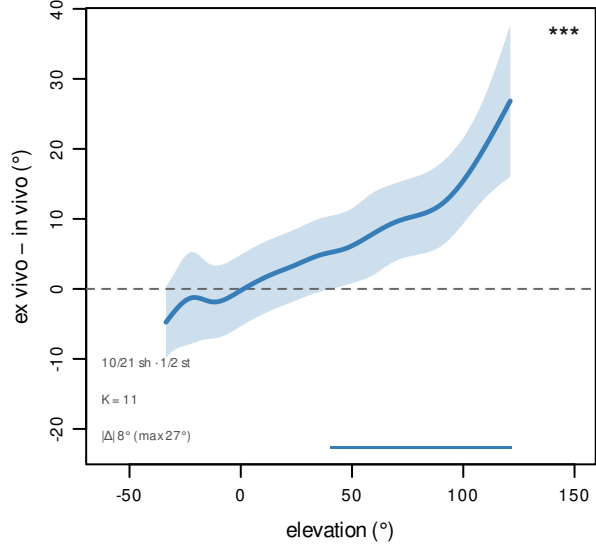
Acromioclavicular — Medial(+)/lateral(-) rot

single group · K = 17

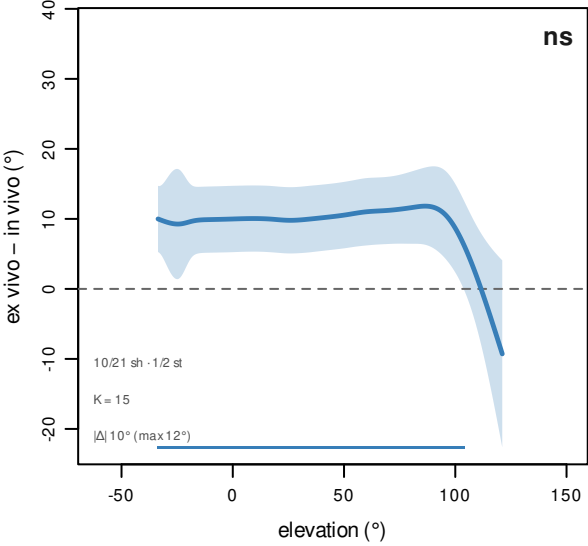
Acromioclavicular — Posterior(+)/anterior(-) tilt

single group · K = 17

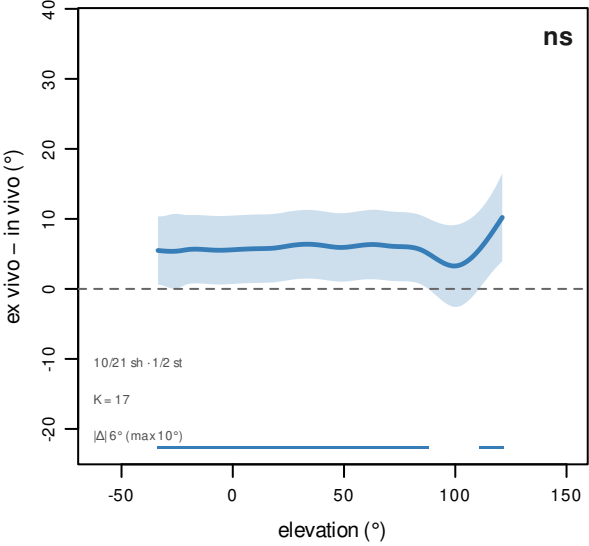
Scapulothoracic — Protraction(+)/retraction(-)



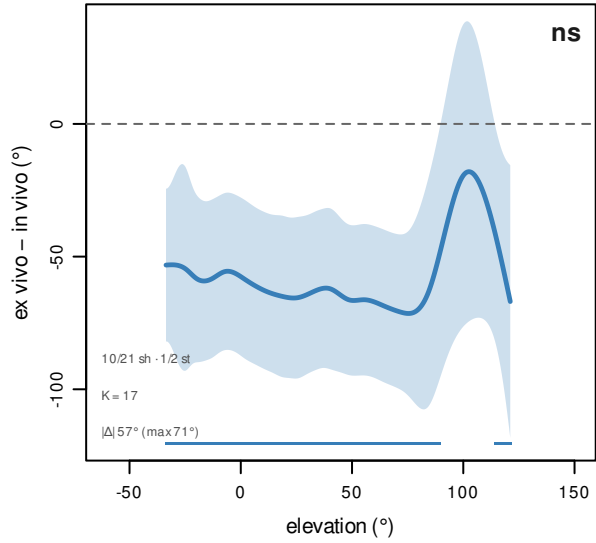
Scapulothoracic — Medial(+)/lateral(-) rot



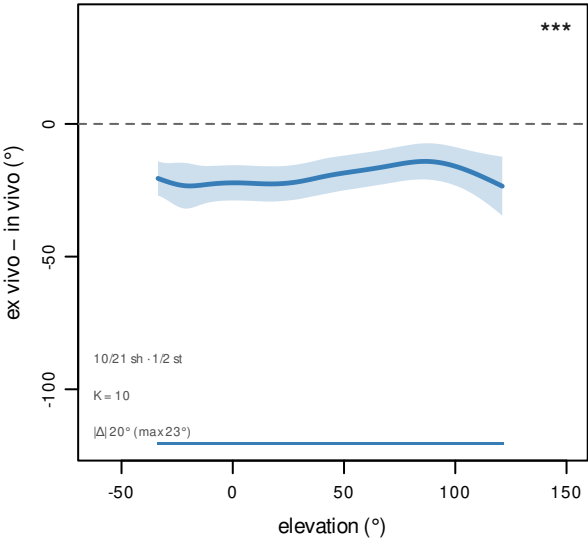
Scapulothoracic — Posterior(+)/anterior(-) tilt



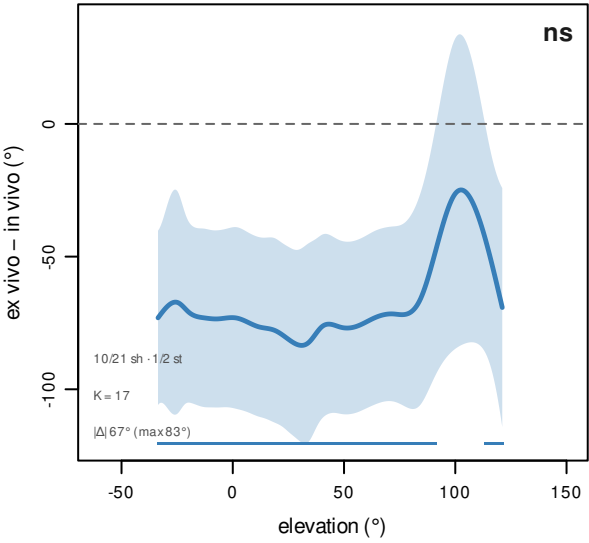
Glenohumeral — Plane of elevation



Glenohumeral — Elevation(-)/Depression(+)



Glenohumeral — Int(+)/ext(-) rotation



blue line = EX VIVO – IN VIVO, band = 95% CI · squares along the bottom = band excludes 0. Note the sign: in vivo is the reference, so this is mirrored against v2.

stars = LIKELIHOOD-RATIO TEST of H0 'no ex-vivo departure' against H1, Bonferroni-adjusted across every compared cell on the plate: *** p<0.001, ** p<0.01, * p<0.05, ns = not significant.

ex/in · studies ex/in | K, this cell's own spline df chosen by BIC | mean and max |Δ| in degrees. Curves span the full abscissa — every x any shoulder reaches is fitted, so the thinnest stretches rest on a single shoulder and the 95%